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Roll No_____

FIFTH SEMESTER

B.Tech. [ALL]

MID SEMESTER EXAMINATION

September-2019

IT-351 ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Time: 1 Hour and 30 Minutes

Max. Marks: 25

Note: Answer *ALL* questions.
Assume suitable missing data, if any.

Question No. 1

[1x5=5]

Translate in two ways each of these statements into logical expressions using predicates, quantifiers, and logical connectives. First, let the domain consist of the students in your class and second, let it consists of all people.

[a] Someone in your class can speak Hindi.

[b] Everyone in your class is friendly.

[c] There is a person in your class who was not born in California.

[d] A student in your class has been in a movie.

[e] No student in your class has taken a course in logic programming.

Question No. 2

[3+2=5]

[a] What are the challenges in AI? Explain how utility-based agents help to improve the performance of AI systems.

[b] Consider, there are two water Jars, and one measures 5 Gallons (5G) while the other measures 7 Gallons (7G). There is no measuring label mentioned on either of these two Jars i.e. the exact amount of water only be known when Jar is completely filled. Assume that there is a countless amount of water supply, explain how one can measure 1G and 3G using these unmarked Jars by representing it as a state-space problem. (You can empty the Jars, and transfer water from one Jar to other).

Question No. 3

[5]

Consider a Tree graph as given in Figure No. 1, apply an alpha-beta pruning approach to obtain optimal solution.

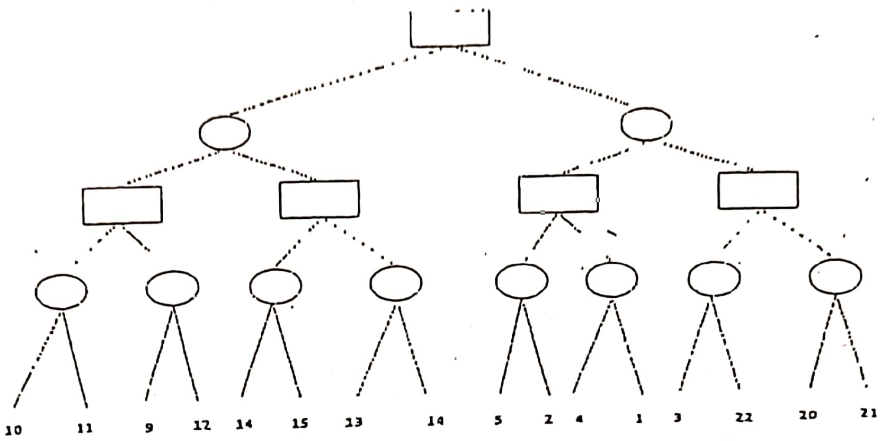
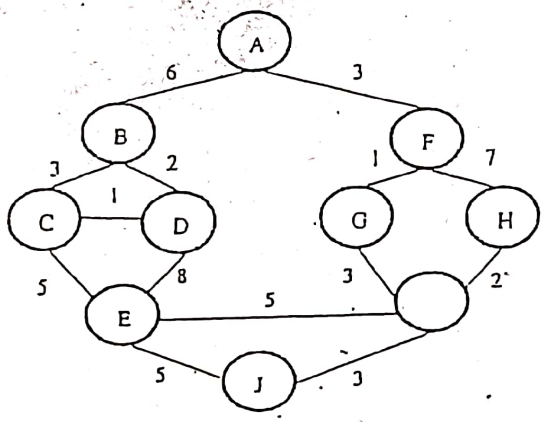


Figure No. 1

Question No. 4 [5]

Describe the Heuristic and Greedy terms in artificial intelligence. Find the shortest path from vertex A to vertex J using A* algorithm.



Node	Heuristic Value
A	10
B	8
C	5
D	7
E	3
F	6
G	5
H	3
I	1
J	0

Question No. 5 [3+2=5]

[a] Explain how hill climbing is a greedy local search-based optimization algorithm. Discuss the challenges in hill climbing.

[b] Identify the differences between artificial intelligence, machine learning, and deep learning.